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Exercise 2 : SQL foundations
Aggregate functions & operators

1. all distinct departments in students table

```
SELECT DISTINCT departments
FROM students;
```

Department
IT
HR
FINANCE

2. get average age of students per department.

```
SELECT department,
       avg (age) AS avg-age
FROM students
WHERE age AS Avg-age
GROUP BY department ;
```

Department	avg-age
IT	20.5
HR	22
FINANCE	23

③ Show departments with more than 1 student.

```
SELECT department,
       student count AS (COUNT(*))
FROM Students
GROUP BY department
HAVING COUNT (*) > 1;
```

department	student count
IT	2
HR	2

④ Get all students whose age is between 21 and 23

```
SELECT student ID,
       name,
       age,
       department
  FROM Students
WHERE AGE BETWEEN 21 AND 23;
```

Student ID	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

(5) list all student in the IT or HR depotment who are older than 21

```
SELECT student ID ,
       name,
       age,
       department
  FROM students
WHERE ~~GROUP BY~~ Department = 'IT' OR 'HR AND
  Age >21
```

student ID	name	age	Department
2	Bob	22	HR
3	Challie	21	IT
5	Eve	22	HR

(6) Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department ,
       Total credits
  FROM ~~students~~ courses
  GROUP BY department
  ~~WHERE Total credits >5;~~
  HAVING sum (credits >5;
```

Department	total credits
IT	11

7. list all courses that do not have 4 credits.

```
SELECT  course ID, ~~co~~
        course name,
        department,
        credits
FROM    ~~students~~ courses
WHERE   credits <> 4;
```

course ID	course name	department	credits
101 ~~SQL Basics~~	SQL Basics	IT	3
104	Excel	finance	2
105	Statistics	HR	3

8. show the top 3 courses by credits in descending order

```
SELECT  course_ID,
        course_ name,
        credits
FROM    courses
ORDER BY credits desc
Limit 3;
```

course_ID	course name	credits
102	Python	4
103	Data Science	4
105	Statistics	3

⑨ Get maximum, minimum & average grade across all enrollments

```
SELECT max(grade) AS Max_grade,
       min(grade) AS MIN-grade,
       avg(grade) AS AVG_grade
FROM enrollments;
```

max_grade	Min grade	Avg-grade
90	78	84.6

⑩ Count how many enrollments exist per course.

```
SELECT course ID,
  COUNT(*) AS enrollment_COUNT
FROM enrollments
GROUP BY course_ID;
```

course ID	enrollment_count
101	1
102	1
103	1
104	1
105	1

⑪ find total salary & total bonus per department

```
SELECT Department,
       Sum(Salary) AS total_salary,
       sum(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

Department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
finance	70 000	6000

(12) Show departments where Average salary is above 55000

```
SELECT department,
    avg (salary) AS Avg_salary
  FROM Salaries
WHERE Avg-Salary > 55 000 $
GROUP BY department;
```

Department	avg_salary
IT	61,000
~~HR~~	~~54 500~~
finance	70000

Department	Avg_salary
IT	61 000
finance	70 000

(13) list employees whose salary plus bonus is greater than 60 000

```
SELECT employee_ID;
       name,
       salary,
       bonus,
sum (compensation) AS total_compensation
 FROM Salaries
 ~~WHERE~~ HAVING total_compensation > 60000
```

employee-ID	name	salary	bonus	total_compensation
5	Butch	$4000	3500	57500

(14) Show total and average budget per department. Only include departments with average budget above 70 000.

```
SELECT department,
sum (budget) AS total_budget,
avg (budget) As avg_budget
FROM Projects
GROUP BY department
Having avg_budget > 70000;
```

Department	total_budget	avg_budget
IT	270 000	135 000
finance	80 000	80 000

(15) list all projects with budgets between 50000 and 120000, excluding the marketing department.

```
SELECT project_ID,
       Project_name,
       department,
       budget
FROM Projects
WHERE Department <> 'Marketing' ~~AND~~
HAVING budget BETWEEN 50 000 AND 120 000
```

Project ID	Project name	Department	budget.
1	AI APP	IT	120 000
2	Payroll systems	finance	80 000
5	HR portal	HR	50 000